



News and Geolocated Social Media Accurately Measure Protest Size Variation

Sobolev et al. (2020)

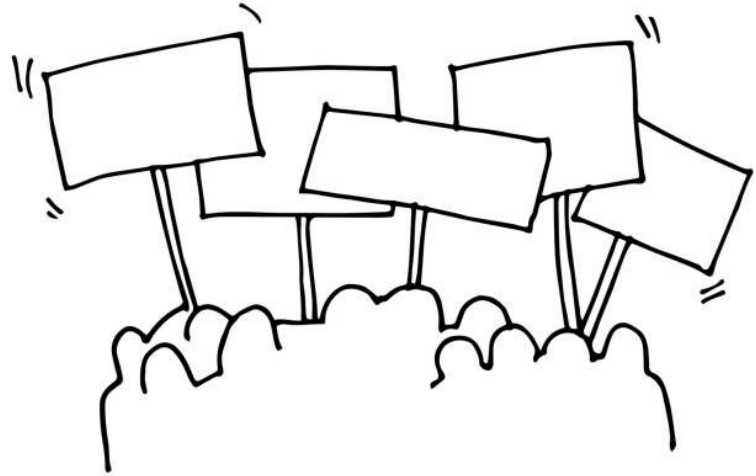
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Computational Social Science with Images & Audio
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Motivation

- Protests draw attention to issues otherwise ignored by policymakers.
- They may also signal that a social movement is organized and persistent.
- Larger protests are more likely to lead to policy changes than small ones are (Wouters & Walgrave, 2017)

But how accurate are their attendance estimates?



Research Question



- **Are attendance estimates provided in news or generated from social media biased?**
- *News reports rely on others' estimates, and the social media estimates are from models applied to relatively small samples with many entry points for bias. That both sets of estimates work may therefore surprise (pp. 1349).*
- **Sources of bias:** interest to misrepresent protest size, vagueness of information, coverage and reporting bias, lack of internally valid of estimates.

Approach

- Compare the accurateness of several measurement approaches against a ground truth: *cellphone data*
- The cellphone data shows that variations in protest size reported in newspapers, as well as estimated via geolocated tweets, are accurate.



Cellphone data



CCC: News

CCC: All Estimates (Best Guess)

CCC: All Estimates (Low)



Twitter: Text Accounts

Twitter: Images Faces

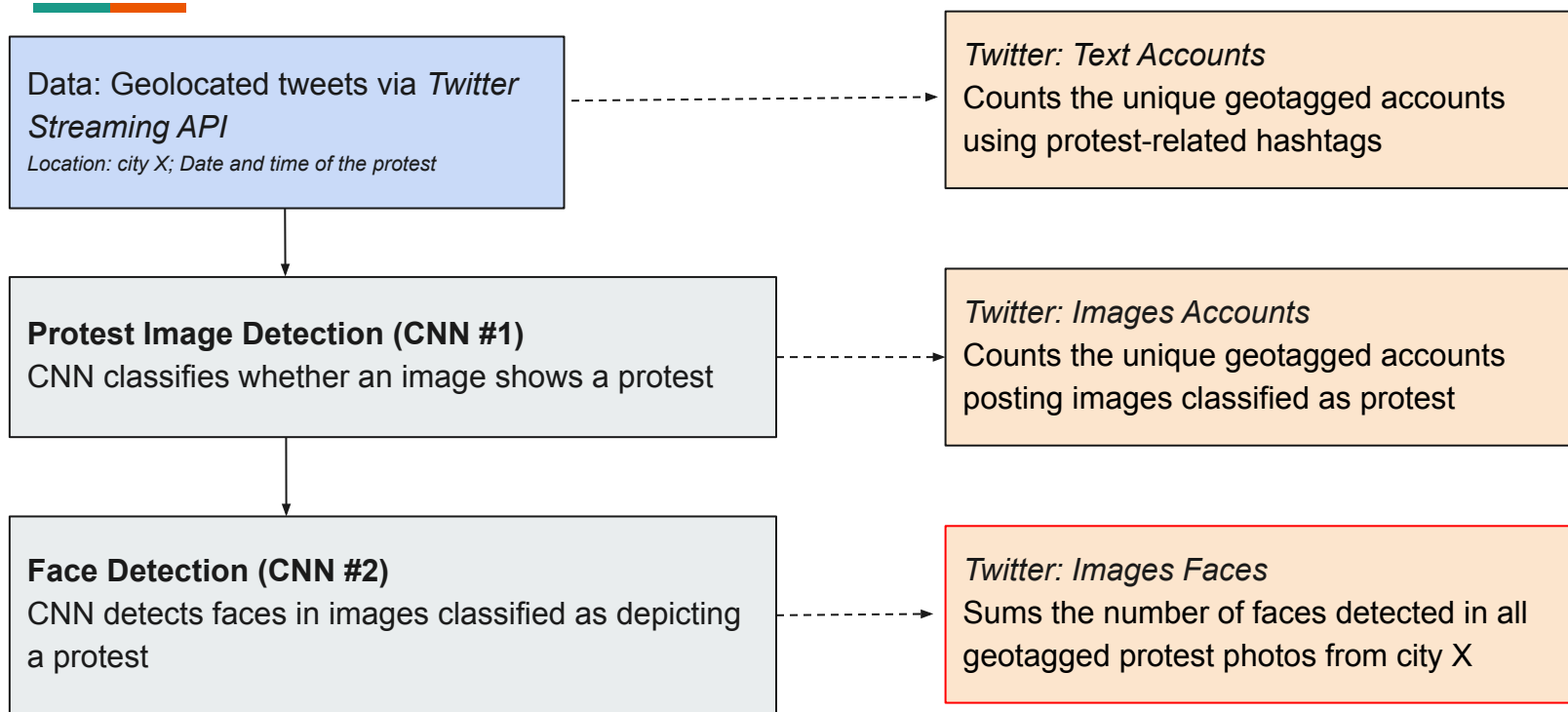
Twitter: Images Accounts

Case study

- That news and geolocated social media measure protest size variation during the **2017 United States' Women's March**.
- Data of protest events from **155 cities** was used to produce the results.
- Cities that have more people, higher earners, greater income inequality, and more inhabitants with a professional degree are more likely to record protests.



Pipeline - Twitter Measures



Twitter: Images Faces - Details



Idea: Each geotagged protest photo = a small **visual sample** of the crowd

→ Many photos, posted randomly across the protest

→ **Sum of detected faces $\approx$ protest size variation**

Implementation: uses the **pre-trained CNNs** developed in *Steinert-Threlkeld et al. (2020)*:

- CNN #1: a classifier distinguishing protest vs. non-protest images
- CNN #2: a face-detection model (based on the *FairFace*)

Robustness check: duplicate faces are very rare (<1%)

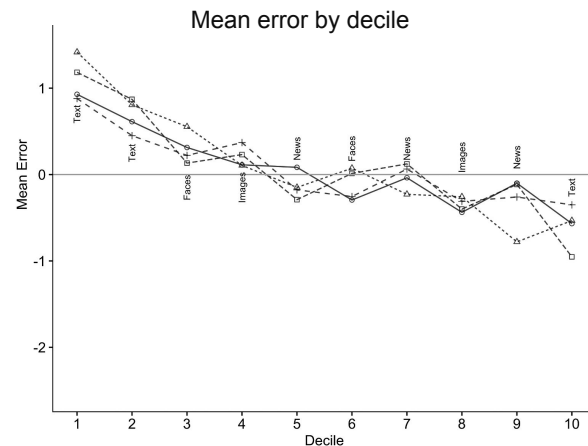
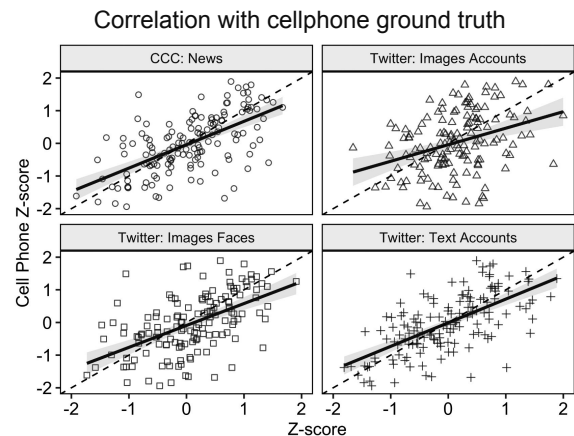
Results

Core result: News reports and geolocated Twitter measures **accurately capture variation** in protest size when benchmarked to cellphone location data

Different measures perform differently:

- **Best:** *CCC: News* and *Twitter: Text Accounts*
- **Weakest:** *Twitter: Images Accounts*

Systematic error pattern: overestimate small protests and underestimate large ones



Strengths & Limitations



Strengths

Multimodal Validation

High applicability useful in contexts with scarce or unreliable protest data (e.g., authoritarian regimes)

Automated & Scalable: once built, pipeline runs quickly and cheaply across contexts

Limitations

Relative, not absolute size

Generalizability: tested only on the U.S. Women's March – unusually large, peaceful, and well-documented. In in sensitive or unauthorized protests participants may avoid posting identifiable faces.

Sampling: few tweets are geotagged (<1%) (Kruspe et al., 2021), limites sample size and representativeness

Platform dependency: reliant on Twitter API and policy stability

Implications:



Research:

- Validated news-based counts capture relative turnout → we can use media data to test what drives mobilization and how cross-city size differences lead to divergent societal effects.
- Even in hard-to-reach places (authoritarian contexts, conflict settings) where other information is scarce or censored, the paper provides a smart approach to estimate protest size.

Society:

- The same method can be used for example to study **state repression** in the context of protests. (Steinert-Threlkeld et al. 2022)

Ethics:

- **Risk:** Protest photos on social media can help authorities identify and arrest people—faces are visible, the place/time is often obvious from the scene, and copies keep circulating even if the original is deleted.
- **Safer practice:** For research use of protest images, do not retain original files or any personal identifiers; refrain from any identification efforts; and release only delayed, city-level, de-identified aggregates. In high-risk contexts, forgo imagery altogether and rely on independent news/NGO tallies or aggregate city-day text counts.



Discussion

1. What ethical challenges arise when researchers rely on data from private platforms whose methods and policies are beyond academic control?

2. Do protesters strategically choose what to post (or not post), and how might this behavior systematically bias image-based or hashtag-based measures?



References



Kruspe, A., Häberle, M., Hoffmann, E. J., Rode-Hasinger, S., Abdulahhad, K., & Zhu, X. X. (2021). Changes in Twitter geolocations: Insights and suggestions for future usage. *arXiv preprint arXiv:2108.12251*.

Sobolev, A., Chen, M. K., Joo, J., & Steinert-Threlkeld, Z. C. (2020). News and Geolocated Social Media Accurately Measure Protest Size Variation. *American Political Science Review*, 114(4), 1343–1351.

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Wouters, Ruud, and Kirsten Van Camp. 2017. “Less than Expected? How Media Cover Demonstration Turnout.” *The International Journal of Press/Politics* 22 (4): 450–70.